COMPOSITION OF WEIGHTINGS FOR LOCALLY QUASI-FINITE MORPHISMS

THE STACKS PROJECT AUTHORS

Editorial note. Counted span is the author's own proof environment, source lines 22261-22309, immediately following the statement at 22249-22259.

Editorial note. The proof cites three chapter lemmas by \ref; these are navigation pointers to standing inputs of the surrounding chapter, so their statements are not re-counted here.

Editorial note. The closing line 'It follows immediately that ...' is retained verbatim: it only multiplies the two constant values already produced inside the span.

Allowed standard prerequisites (editorial).

- Criterion for a weighting after etale base change. chapter lemma lemma-weighting-check-after-etale-base-change, cited at proof lines 22273-22276. Lets the local constancy of the integral be checked on an etale neighbourhood of a point of the base.
- Etale splitting off of the quasi-finite part. chapter lemma lemma-etale-splits-off-quasi-finite-part-technical-variant, cited at proof lines 22280-22281. Decomposes the finite part over an etale neighbourhood into pieces each having a unique point with purely inseparable residue extension over the base point.
- Quasi-finiteness becomes finiteness after an elementary etale neighbourhood. chapter lemma lemma-etale-makes-quasi-finite-finite-at-point, cited at proof line 22293. Produces an open neighbourhood of the image of the marked point over which the base change is finite, yielding the intermediate finite morphism V -> W.
- Etale morphisms are open. standard Stacks fact used at proof line 22274. Justifies replacing the base by an etale neighbourhood when testing local constancy.

Source reference. Original lemma 02LK: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 11924–11924. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original lemma 02LO: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 12136–12136. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original lemma 0F39: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 22111–22111. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted

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proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Lemma 0.1. *Let $f : X \to Y$ and $g : Y \to Z$ be locally quasi-finite morphisms. Let $w_f : X \to \mathbf{Z}$ be a weighting of f and let $w_g : Y \to \mathbf{Z}$ be a weighting of g. Then the function*

$$X \longrightarrow \mathbf{Z}, \quad x \longmapsto w_f(x)w_g(f(x))$$

is a weighting of $g \circ f$.

Proof. Let us set $w_{g\circ f}(x) = w_f(x)w_g(f(x))$ for $x \in X$. Consider a diagram

$$\begin{array}{ccc} X & \longleftarrow & U \\ {\scriptstyle g\circ f}\downarrow & & \downarrow{\scriptstyle \pi} \\ Z & \longleftarrow & W \end{array}$$

where $W \to Z$ is étale, $U \subset X_W$ is open, and $U \to W$ finite. We have to show that $\int_\pi w_{g\circ f}|_U$ is locally constant. Choose a point $w \in W$. By Lemma 0F39 (and the fact that étale morphisms are open) it suffices to show that $\int_\pi w_{g\circ f}|_U$ is constant after replacing (W, w) by an étale neighbourhood. After replacing (W, w) by an étale neighbourhood we may assume $U = U_1 \amalg \ldots \amalg U_n$ where each U_i has a unique point u_i lying over w such that $\kappa(u_i)/\kappa(w)$ is purely inseparable, see Lemma 02LO. Clearly, it suffices to show that $\int_{U_i \to W} w_{g\circ f}|_{U_i}$ is constant in an étale neighbourhood of w. This reduces us to the case discussed in the next paragraph.

We have $w \in W$ and there is a unique point $u \in U$ lying over w with $\kappa(u)/\kappa(w)$ purely inseparable. Consider the point $v = f(u) \in Y$. After replacing (W, w) by an elementary étale neighbourhood we may assume there is an open neighbourhood $V \subset Y_W$ of v such that $V \to W$ is finite, see Lemma 02LK. Then $f_W^{-1}(V) \cap U$ is an open neighbourhood of u where $f_W : X_W \to Y_W$ is the base change of f to W. Hence after Zariski shrinking W, we may assume $f_W(U) \subset V$. Thus we obtain morphisms

$$U \xrightarrow{a} V \xrightarrow{b} W$$

and $U \to V$ is finite as $V \to W$ is separated (because finite). Since w_f and w_g are weightings of f and g we see that $\int_a w_f|_U$ is locally constant on V and $\int_b w_g|_V$ is locally constant on W. Thus after shrinking W one more time we may assume these functions are constant say with values n and m. It follows immediately that $\int_\pi w_{g\circ f}|_U = \int_{b\circ a} w_{g\circ f}|_U$ is constant with value nm as desired. □